EXPERIMENT 20-

AIM:

To write a C program to demonstrate error propagation in ECB and CBC encryption modes and analyze the effect of transmission errors on plaintext and ciphertext blocks.

CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
   char p[5][20], c[5][20], d[5][20];
   int i, j, n;

   printf("Enter number of blocks: ");
   scanf("%d", &n);

   printf("Enter plaintext blocks:\n");
   for(i = 0; i < n; i++)
      scanf("%19s", p[i]);

   // Simple XOR-based demonstration
   for(i = 0; i < n; i++) {
      strcpy(c[i], p[i]);
      for(j = 0; c[i][j]; j++)
         c[i][j] ^= 5;
```

```
    }

    printf("\nCiphertext blocks:\n");
    for(i = 0; i < n; i++)
        printf("C%d = %s\n", i + 1, c[i]);

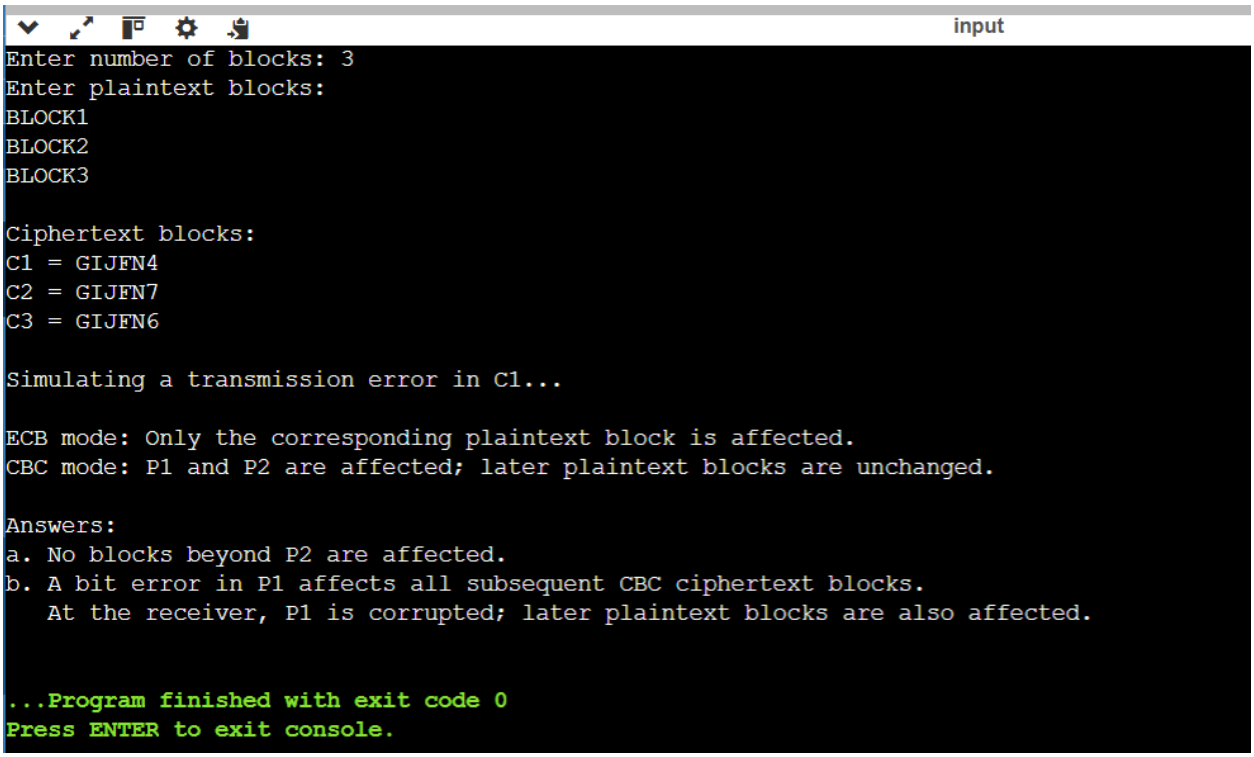
    printf("\nSimulating a transmission error in C1...\n");
    c[0][0] ^= 1;

    printf("\nECB mode: Only the corresponding plaintext block is
affected.\n");
    printf("CBC mode: P1 and P2 are affected; later plaintext blocks are
unchanged.\n");

    printf("\nAnswers:\n");
    printf("a. No blocks beyond P2 are affected.\n");
    printf("b. A bit error in P1 affects all subsequent CBC ciphertext
blocks.\n");
    printf("   At the receiver, P1 is corrupted; later plaintext blocks are also
affected.\n");

    return 0;
}
```

OUTPUT:


input
Enter number of blocks: 3
Enter plaintext blocks:
BLOCK1
BLOCK2
BLOCK3
Ciphertext blocks:
C1 = GIJFN4
C2 = GIJFN7
C3 = GIJFN6
Simulating a transmission error in C1...
ECB mode: Only the corresponding plaintext block is affected.
CBC mode: P1 and P2 are affected; later plaintext blocks are unchanged.
Answers:
a. No blocks beyond P2 are affected.
b. A bit error in P1 affects all subsequent CBC ciphertext blocks.
At the receiver, P1 is corrupted; later plaintext blocks are also affected.
...Program finished with exit code 0
Press ENTER to exit console.